Sahaj Bhadja

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Summary

Software engineer pursuing an MSCS at USC with 2+ years building AI-enabled products, backend services, and debugging workflows across APIs, logs, SQL data, and LLM applications, with strength in root-cause analysis and user-facing communication.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Investigated SQL-backed workflow issues for 500+ users, reducing latency from **1.2s to 350ms** through log review, query tuning, indexed access patterns, and backend fixes.
- Built ETL-style validation pipelines processing 5,000+ monthly jobs with reconciliation checks, CI/CD releases, and clear runbooks that improved triage speed for data and integration failures.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed React and Flask features handling 10,000+ daily requests, debugging API behavior, database states, and user-reported failures while improving usability, tests, and release confidence.
- Built AI-assisted grading workflows with OCR, LLM prompts, and AWS-backed persistence, reducing feedback turnaround by **60%** while documenting edge cases and model-response limitations.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built Python evaluation pipelines across 100,000+ records, producing reproducible traces, reports, and debugging artifacts that clarified model behavior and improved experimental reliability.
- Analyzed failure patterns across generated outputs and datasets, translating technical findings into concise recommendations for researchers while strengthening validation coverage and root-cause communication.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered API and ETL microservices using Python, AWS Glue, and S3, improving response time by **30%** while supporting scalable integrations and reliable production workflows.
- Maintained 20+ automated test suites using Jenkins, PyTest, and Mockito, increasing CI reliability by **68%**, reducing regressions, and accelerating defect isolation across services.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a MERN-based platform with WebSockets and REST APIs, supporting 2M+ synchronized records and six workflows through modular services and responsive user-facing interfaces.
- Automated Docker-based build and test flows with Jenkins, cutting release time by **50%**, improving deployment consistency, and reducing friction during validation and troubleshooting.

Skills

AI & Debugging: LLM prompts, RAG, agent workflows, OCR, failure analysis, logs, telemetry, runbooks, user enablement
Languages: Python, TypeScript, JavaScript, Java, SQL, C++, Bash
Backend & APIs: Flask, FastAPI, Node.js, REST APIs, Postman, microservices, backend integrations, secure services
Data & Delivery: PostgreSQL, MySQL, SQL Server, MongoDB, AWS, Azure DevOps familiarity, Docker, Kubernetes, Git, Jenkins, CI/CD

Projects

Big-Brain Live | *LLaMA-2, FAISS, PostgreSQL, GCP*
- Built a secure document Q&A platform with semantic search, summarization, and prompt workflows over 50K+ records, reducing review time by **60%**.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Built an AI-powered SQL workspace supporting 10+ databases, combining schema inspection, query generation, and debugging flows for faster data validation and root-cause analysis.

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Developed an NLP service pipeline with containerized deployment and CI/CD automation, improving reproducibility, API reliability, and troubleshooting speed for model-backed language workflows.